

# Hollow Soul - Unity

M12 to M25 milestone summary for Rafal

## Purpose

This PDF summarizes the post-M11 phase of Hollow Soul - Unity: what each milestone added, how the prototype architecture changed, what is playable now, and where Rafal should look first when reviewing the project.

|                    |                                                                                     |
|--------------------|-------------------------------------------------------------------------------------|
| Prepared for       | Rafal                                                                               |
| Project            | Hollow Soul - Unity                                                                 |
| Repository root    | /Users/martinjedrzejewski/Documents/GitHub/Unity/Hollow Soul - Unity                |
| Milestones covered | M12 through M25                                                                     |
| Date               | April 26, 2026                                                                      |
| Source material    | Repo milestone docs, tests, generated reports, and current implementation direction |

Executive read: M12-M25 converted the early Unity prototype into a repeatable vertical-slice branch pipeline: build automation, macro-room runtime, seeded branch generation, approved designer-room intake, seeded rewards, enemy encounter content, boss-key/secret/shop branch features, shop card UI, macro Room Designer authoring, ArtPass replacement content, platform QA, and a locked vertical slice report.

# Executive Summary

The post-M11 phase moved the project from a validated prototype shell into a branch-ready vertical slice. The key architectural decision stayed consistent: gameplay truth comes from runtime data and controllers, while art, USDA, and authored visuals remain presentation-only.

**Runtime foundation**

Macro rooms now import and render as one logical room instance across 1x1, 2x1, 1x2, 2x2, and L footprints. Branch traversal can use exact door port IDs.

**Procedural branch layer**

New runs progressed from fixed macro branches to seeded eight-room macro trees, then feature branches with treasure, random rewards, encounter planning, secrets, boss-key locks, and an inter-branch hub.

**Authoring pipeline**

Room Designer evolved into a macro-room authoring workflow with validation, preset footprints, semantic tools, runtime JSON export, and approval into the branch room pool.

**Content and QA**

ArtPass prefabs, material/cue catalogs, Addressables labels, platform QA profiles, and a vertical-slice lock gate now define a repeatable handoff surface.

## Current vertical-slice shape

- Start a profile-backed New Run into the branch-feature runtime path.
- Clear seeded combat rooms, collect automatic rewards, find/use the boss key, enter the optional visible secret, clear the boss room, enter the inter-branch hub, buy a shop card, and inspect next-branch portals.
- Windows, Vision Pro bounded tabletop, and Vision Pro immersive routes share gameplay logic and differ through presentation profiles, world scale, and QA checklist expectations.

## Recent cleanup note

After M25, presentation/UX fixes continued: Room Designer entry flow, rock ground alignment, shell overlay removal, HUD separation, and gameplay camera follow. These are cleanup patches around the milestone lock rather than new formal milestone identities.

## Milestone Map: M12 to M18

| Milestone | Theme                         | What changed                                                                                         | Why it matters                                                                                     |
|-----------|-------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| M12       | Build automation and audit    | Full prototype audit runner, build manifest, Windows development build endpoint, report conventions. | Makes the project repeatable for another engineer and creates the first build/report gate.         |
| M13       | Macro-room foundations        | Imported/designer rooms preserve 1x1, 2x1, 1x2, 2x2, and L footprints as one logical room instance.  | Prepares branch generation for larger room shapes without splitting rooms into fake smaller nodes. |
| M14       | Seeded macro fixture branch   | New Run uses a fixed five-room macro branch with explicit port-to-port connections.                  | Proves macro room placement and traversal while preserving save/reward compatibility.              |
| M15       | Seeded procedural branch v1   | Eight-room seeded macro tree, deterministic rewards, one boss leaf, exact door ports.                | Moves from fixture topology to actual deterministic branch generation.                             |
| M16       | Approved designer room pool   | Repo-tracked DesignerApproved intake folder and validation for exported Room Designer runtime JSON.  | Lets authored rooms become branch candidates without destabilizing fallback fixtures.              |
| M17       | Feature branch treasure rooms | Adds one special treasure leaf, auto-clear behavior, and Treasure Cache reward.                      | Introduces non-combat room roles while keeping branch generation stable.                           |
| M18       | Seeded random rewards         | Role-based seeded reward pools replace fixed reward cycling and persist exact reward plans.          | Rewards become random-by-seed, automatic, and continue-safe.                                       |

## Milestone Map: M19 to M25

| Milestone | Theme                      | What changed                                                                                                                            | Why it matters                                                                    |
|-----------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| M19       | Enemy encounters V1        | Enemy behavior IDs, charger/turret/splitter, boss warden behavior, seeded encounter catalog/plans.                                      | Turns rooms from generic spawns into authored encounter content by seed.          |
| M20       | Branch features            | Boss key, locked boss path, visible secret room, inter-branch hub, shop stand, next-branch portals.                                     | Creates a complete branch progression loop beyond one boss clear.                 |
| M21       | Shop choice UI             | Visible in-world shop cards, card-specific interact purchase, SOLD state, fixed prices.                                                 | Makes shop behavior player-facing and testable rather than hidden auto-buy logic. |
| M22       | Room Designer macro polish | Draft library/create presets, macro visualization, enemy variants, port states, validation-gated export/playtest.                       | Makes room authoring useful for branch-ready content handoff.                     |
| M23       | Art/content replacement    | Presentation prefab roles, ArtPass palette/prefabs/VFX/SFX, validation, local Addressables labels.                                      | Lets visuals improve without giving art authority over gameplay or collision.     |
| M24       | Platform build/device QA   | Platform QA profile, Addressables build gate, Windows dev build path, Vision Pro readiness checks, runtime QA probe.                    | Turns editor validation into a platform handoff gate.                             |
| M25       | Vertical slice lock        | Locks branch identity m20_branch_features_v1 and seed 15001, validates content, platform checklists, ArtPass requirements, reports/PDF. | Defines the first repeatable vertical-slice review target.                        |

## How The Current Slice Fits Together

The useful mental model for Rafal: Hollow now has a data-first runtime pipeline. Room JSON and ScriptableObject catalogs feed deterministic branch/session systems; visuals attach as children and never replace gameplay authority.

|                      |                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Room data         | HollowRuntime V2 JSON from fixtures or approved Room Designer exports. Contains walkable tiles, footprint, door ports, obstacles, safe start, and spawn markers.  |
| 2. Branch generation | Seeded branch generators place macro-room instances, preserve occupied cells, and connect exact door port IDs. Legacy branch paths remain restore-compatible.     |
| 3. Plans by seed     | Reward plans and encounter plans are generated from branch seed/catalog data, serialized into snapshots, and restored exactly on Continue.                        |
| 4. Runtime session   | BranchSessionController loads the current room asset, starts combat or auto-clear roles, routes traversal, pickups, boss key, hub, shop, and next branch portals. |
| 5. Presentation      | RoomRuntimeRoot and combat/reward objects keep colliders/components authoritative, then attach ArtPass visual prefabs/materials through presentation catalogs.    |
| 6. QA/lock           | M24 and M25 generate reports that check platform readiness, ArtPass coverage, branch features, save/continue points, and locked seed content.                     |

### Key rule to preserve

- Do not infer gameplay from visual prefabs, meshes, USDA companions, or ArtPass hierarchy.
- Do keep runtime state plain/serializable and presentation as replaceable adapters/children.
- Do keep transient designer/playtest sessions separate from profile-backed run saves.

## Locked Vertical Slice Route

M25 locks a reviewable route around branch identity m20\_branch\_features\_v1 and seed 15001. The route is designed to verify the core loop and the systems added in M12-M25 without introducing new gameplay at the lock stage.

| Step | Player-facing event                             | Systems exercised                                                                       |
|------|-------------------------------------------------|-----------------------------------------------------------------------------------------|
| 1    | Create/select profile and start New Run.        | Profile/session route, active run snapshot, branch seed and lock config.                |
| 2    | Enter origin/combat rooms and clear encounters. | Room import, macro footprint rendering, enemy encounter plan, combat loop, checkpoints. |
| 3    | Collect automatic room rewards.                 | Seeded reward plan, run-local souls, stat effects, persistence.                         |
| 4    | Find boss key source room and collect boss key. | Feature plan, key pickup, reward replacement, boss-key state save.                      |
| 5    | Optionally enter visible secret room.           | Secret door gating, no-combat auto-clear, optional bonus reward.                        |
| 6    | Unlock boss door and defeat Stone Warden.       | Locked connection, boss encounter profile, boss reward.                                 |
| 7    | Enter inter-branch hub.                         | Boss portal, in-run hub state, no meta banking.                                         |
| 8    | Buy a specific shop card.                       | M21 card UI, fixed-price purchase, SOLD persistence, immediate effect.                  |
| 9    | Inspect three next-branch portals.              | Seed derivation for branch continuation and portal presentation.                        |

Important: M25 intentionally locks content/readiness rather than adding systems. It is the review target, not the end of feature development.

## Where Rafal Should Look First

| Area                | Useful files/folders                                       | What to check                                                                                                    |
|---------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Milestone docs      | Docs/Milestone12...25*.md                                  | Narrative intent, commands, scope boundaries, and validation expectations.                                       |
| Branch/runtime      | Assets/_Hollow/Scripts/Hollow.Branches/ and Hollow.Rooms/  | Macro footprint handling, explicit port traversal, feature plan, room asset selection, key/secret/hub/shop flow. |
| Combat/enemies      | Assets/_Hollow/Scripts/Hollow.Combat/                      | Encounter startup, enemy behavior IDs, projectile ownership, boss warden behavior.                               |
| Rewards/persistence | Hollow.Rewards/ and Hollow.Persistence/                    | Reward plan save/restore, run-local souls, active run snapshot safety, legacy Continue paths.                    |
| Room Designer       | Hollow.RoomDesigner/ and Data/Rooms/DesignerApproved/      | Preset creation, validation rules, runtime JSON export, manual approval workflow.                                |
| Presentation/art    | Hollow.Presentation/, Data/Presentation/, Prefabs/ArtPass/ | Prefab role bindings, visual-only constraints, material/cue fallbacks, no gameplay collider takeover.            |
| QA/reports          | output/reports/ and output/pdf/                            | Latest audit, platform QA, vertical slice lock report, environment blocks vs real failures.                      |

### Quick validation commands

```
/Applications/Unity/Hub/Editor/6000.4.1f1/Unity.app/Contents/MacOS/Unity -batchmode -projectPath "<repo>" -quit -executeMethod Hollow.Editor.Validation.Milestone25Validator.Validate
```

```
/Applications/Unity/Hub/Editor/6000.4.1f1/Unity.app/Contents/MacOS/Unity -batchmode -projectPath "<repo>" -quit -executeMethod Hollow.Editor.Build.VerticalSliceLockRunner.RunM25LockGate
```

Note: batchmode will fail if the same Unity project is already open in the editor. Close Unity first for CI-style runs.

## Open Risks And Suggested Next Phase

The architecture is now rich enough that the highest-value work is less about adding another mechanic and more about turning the vertical slice into a stable, readable, reviewable play experience.

| Priority | Risk / gap                                                                                                     | Recommended next action                                                                                                         |
|----------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| High     | Presentation readability can regress as branch HUD, item logs, minimap, shop cards, and debug text accumulate. | Keep HUD panels separated, define safe areas for Windows and Vision Pro, and add screenshot-based smoke checks where practical. |
| High     | Room Designer is powerful but needs a clear user-facing library/create/edit loop.                              | Continue polishing draft entry flow, template cards, validation feedback, and camera/cursor framing for macro rooms.            |
| High     | ArtPass exists as a replacement layer, but content is still placeholder.                                       | Replace the most visible roles first: player, rocks/floor/doors, three enemy variants, boss, rewards, shop cards, portals.      |
| Medium   | Procedural branches are deterministic but still content-light.                                                 | Add room-pool variety, encounter weights by depth, and branch pacing rules before expanding room count.                         |
| Medium   | Vision Pro readiness is simulator/checklist-oriented.                                                          | Schedule real bounded tabletop and immersive device passes once signing/build environment is ready.                             |
| Medium   | Persistence is feature-rich and needs ongoing regression coverage.                                             | Keep snapshot compatibility tests for M7, M14, M15, M17, M18, M19, M20, and current lock path.                                  |

### Suggested next phase headline

- Make the locked vertical slice feel intentional: final pass on camera, readable HUD, room designer UX, ArtPass replacements, encounter pacing, and real platform QA.
- Then expand content breadth: more approved designer rooms, more enemy encounter tables, richer shops, secrets hidden behind real discoverability, and branch length/pacing progression.



## Appendix: Milestone Deliverables Reference

| Milestone | Primary doc                                         | Core validation/test surface                         |
|-----------|-----------------------------------------------------|------------------------------------------------------|
| M12       | Docs/Milestone12BuildAutomation.md                  | Build automation tests and boot smoke PlayMode tests |
| M13       | Docs/Milestone13MacroRooms.md                       | Macro room import, ports, designer defaults          |
| M14       | Docs/Milestone14MacroBranchGeneration.md            | Macro branch graph and explicit traversal            |
| M15       | Docs/Milestone15SeededProceduralBranches.md         | Seeded generation, boss leaf, reward plan            |
| M16       | Docs/Milestone16ApprovedDesignerRoomPool.md         | Approved designer room import/catalog tests          |
| M17       | Docs/Milestone17FeatureBranchTreasureRooms.md       | Feature branch and treasure role tests               |
| M18       | Docs/Milestone18SeededRandomRewards.md              | Seeded reward pool/save restore tests                |
| M19       | Docs/Milestone19EnemyEncounterContent.md            | Enemy behavior and encounter plan tests              |
| M20       | Docs/Milestone20BranchFeaturesShopsSecretsKeys.md   | Key/secret/hub/shop branch feature tests             |
| M21       | Docs/Milestone21ShopChoiceUi.md                     | Shop card UI/purchase persistence tests              |
| M22       | Docs/Milestone22RoomDesignerMacroAuthoringPolish.md | Room Designer authoring/validation/export tests      |
| M23       | Docs/Milestone23ArtContentReplacementPipeline.md    | ArtPass/prefab role/content validation tests         |
| M24       | Docs/Milestone24PlatformBuildDeviceQA.md            | Platform QA profile and PlayMode scene smoke         |
| M25       | Docs/Milestone25VerticalSliceContentLock.md         | Vertical slice lock tests and report generation      |

Generated from local repo context. This summary is intentionally written as a handoff guide, not a complete technical specification. For exact implementation details, use the milestone docs and tests listed above.